SolarWinds Threat Detection Using Network Logs

Overview:
This project investigates potential exposure to the SolarWinds cyberattack by analyzing network proxy logs and comparing them with known Indicators of Compromise (IOCs). The goal is to identify internal systems that may have communicated with malicious infrastructure.

Data Used:
- SolarWinds IOC Dataset (known malicious IP addresses)
- Network Proxy Log Dataset (internal network activity)

Tools:
- Splunk: Used to search and correlate network log data
- VirusTotal: Used to validate whether matched IP addresses are malicious

Key Findings:
- Multiple internal systems communicated with IPs found in the SolarWinds IOC dataset
- Example match:
  - IP: 5.252.177.25
  - System: LN-SolarStrike-14
  - Timestamp: 2024-03-05 07:11:28
- VirusTotal confirmed that these IPs are flagged as malicious by multiple vendors

Impact:
The presence of these matches suggests that internal systems may have communicated with suspicious or malicious infrastructure, indicating a potential security risk that requires further investigation.

Recommended Actions:
- Block identified malicious IP addresses at the firewall level
- Investigate affected systems for signs of compromise
- Improve monitoring and detection of suspicious outbound connections

Conclusion:
By combining internal log analysis with external threat intelligence, this project demonstrates how organizations can detect and respond to potential threats more effectively.